CALCULUS · ANTIDERIVATIVES AND DEFINITE INTEGRALS

Riemann Sum Lab

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Riemann Sum Lab** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. For $f(x) = \frac{x^2}{4} + 1$ on $[0, 4]$, will left-endpoint rectangles overestimate or underestimate the area? Why?

   Underestimate, because f is increasing: each left height is the lowest on its interval.

2. Can a definite integral be 0 even when the graph encloses area?

   Yes, if area below the x-axis cancels area above it.

2 Explore use the tool

1. $\frac{x^2}{4} + 1$ on $[0, 4]$: record the Left sums for n = 4, 20 and 100, and the exact integral.

   7.5, 8.94, 9.2536; exact $\frac{28}{3} \approx 9.3333$

2. Same function and interval: record the Midpoint sum for n = 4. Compare it with the Left sum for n = 20.

   9.25. With only 4 rectangles it is closer to 9.3333 than the Left sum with 20 (8.94).

3. sin x on $[-3, 3]$: record the exact integral and the total area.

   Integral 0; total area ≈ 3.98

4. $4 - x^2$ on $[2, 4]$: what is the sign of the integral, and why?

   Negative: $-\frac{32}{3} \approx -10.67$, because the graph is below the x-axis there.

3 Record use the tool and fill in the table

Sums and integrals

f(x)	Interval	Method, n	Sum	Exact integral	Over or under?
$\frac{x^2}{4} + 1$	$[0, 4]$	Right, 4	11.5	$\frac{28}{3} \approx 9.33$	over
$4 - x^2$	$[0, 2]$	Left, 4	6.25	$\frac{16}{3} \approx 5.33$	over
$4 - x^2$	$[0, 2]$	Midpoint, 4	5.375	$\frac{16}{3} \approx 5.33$	over
$\frac{x^3}{8} - x$	$[-2, 2]$	Trapezoid, 4	0	0	equal

4 Explain what did you discover?

1. A Riemann sum adds f(sample point) × Δx for each subinterval.

2. As n increases, the sums approach the definite integral.

3. Rectangles below the x-axis count as negative, so the definite integral measures signed area.

4. For an increasing function, left sums **underestimate** and right sums **overestimate**.

5. Explain the difference between $\int_{-3}^{3} \sin x \, dx$ and the total area between sin x and the x-axis on $[-3, 3]$.
   The integral is 0: the part below the axis (about −1.99) cancels the part above (about +1.99). The total area adds them as positive amounts: about 3.98.

5 Apply a new situation, no tool

1. Estimate $\int_{0}^{6} f(x) \, dx$ with a Right sum and n = 3 if f(2) = 4, f(4) = 7 and f(6) = 9.
   $2(4 + 7 + 9) = 40$

2. A particle has velocity v(t) = 3 − t m/s for $0 \le t \le 5$. Find its displacement $\int_{0}^{5} v(t) \, dt$ and the total distance it travels.
   Displacement 4.5 − 2 = 2.5 m; total distance 4.5 + 2 = 6.5 m.

3. Write $\lim_{n\to\infty} \sum_{i=1}^{n} \left(2 + \frac{3i}{n}\right)^2 \cdot \frac{3}{n}$ as a definite integral.
   $\int_{2}^{5} x^2 \, dx$

Teacher notes

Intended discovery: Riemann sums approach the definite integral as n grows; areas below the axis are negative, so the integral is signed area, and left/right sums under- or overestimate according to whether f increases or decreases.

Likely misconceptions:

- The integral is always the total (positive) area.
- More rectangles is the only way to improve accuracy (midpoint vs. left).
- Left sums always underestimate (only for increasing f).

Answers that can vary: Values students read from the tool may differ in the last decimal place; Try this 2 and 4 accept several intervals.